Network Designing & Management

IT3010

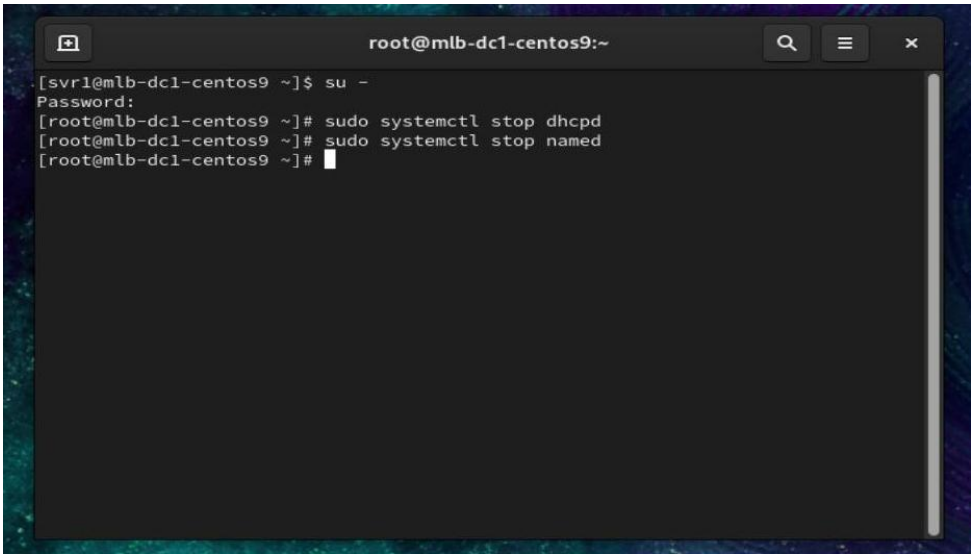
Name: Wickramaarachchi W A C J

Student ID: IT23699076

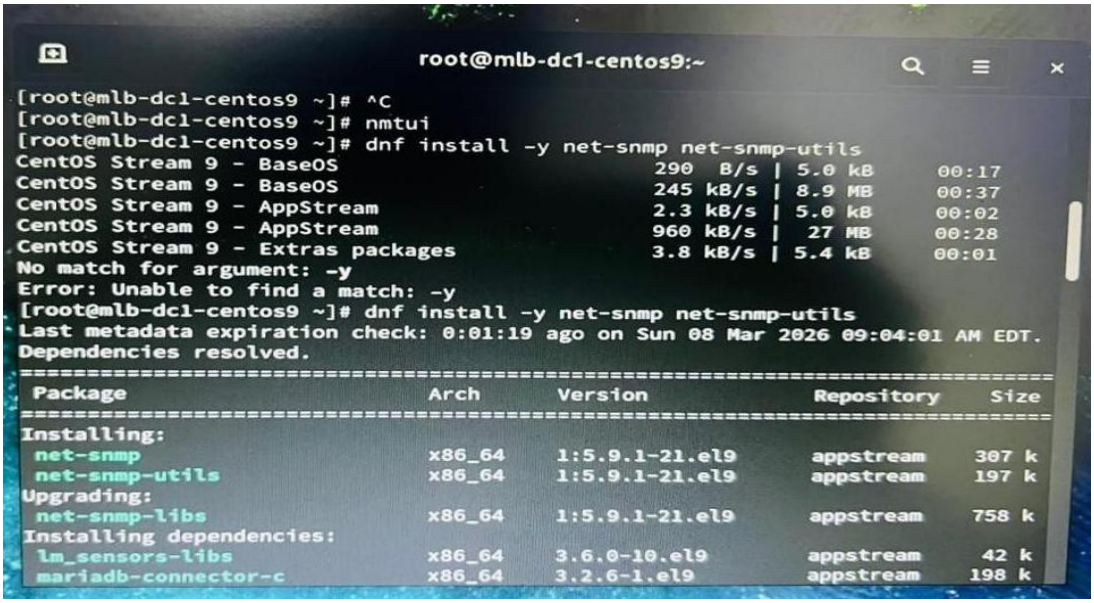
Group: IT_WE 2.2

Practical Number: 05

1) Stop all the running services.



2) Install SNMP using the following command.



3) Use the following command to view the SNMP configuration file.

```
root@mlb-dc1-centos9:~
[root@mlb-dc1-centos9 ~]# sudo systemctl status snmpd
○ snmpd.service - Simple Network Management Protocol (SNMP) Daemon.
     Loaded: loaded (/usr/lib/systemd/system/snmpd.service; disabled; preset: d>
     Active: inactive (dead)

[root@mlb-dc1-centos9 ~]# ^C
[root@mlb-dc1-centos9 ~]# vi /etc/snmp/snmpd.conf
[root@mlb-dc1-centos9 ~]# sudo systemctl start snmpd
[root@mlb-dc1-centos9 ~]# sudo systemctl status snmpd
● snmpd.service - Simple Network Management Protocol (SNMP) Daemon.
     Loaded: loaded (/usr/lib/systemd/system/snmpd.service; disabled; preset: d>
     Active: active (running) since Sat 2026-03-14 01:43:21 EDT; 30s ago
   Main PID: 3703 (snmpd)
      Tasks: 1 (limit: 22774)
     Memory: 9.9M (peak: 10.3M)
        CPU: 137ms
     CGroup: /system.slice/snmpd.service
             └─3703 /usr/sbin/snmpd -LS0-6d -f

Mar 14 01:43:21 mlb-dc1-centos9.csa.lk systemd[1]: Starting Simple Network Mana>
Mar 14 01:43:21 mlb-dc1-centos9.csa.lk snmpd[3703]: NET-SNMP version 5.9.1
Mar 14 01:43:21 mlb-dc1-centos9.csa.lk systemd[1]: Started Simple Network Manag>

[root@mlb-dc1-centos9 ~]# ^C
```

4) Add below configuration to /etc/snmp/snmpd.conf

```
root@mlb-dc1-centos9:~
# enterprises.ucdavis.255.6 = Gauge: 42
#
# % snmpget -v 1 localhost public .1.3.6.1.4.1.2021.255.5
# enterprises.ucdavis.255.5 = 42
#
# % snmpset -v 1 localhost public .1.3.6.1.4.1.2021.255.1 s "New string"
# enterprises.ucdavis.255.1 = "New string"
#

# For specific usage information, see the man/snmpd.conf.5 manual page
# as well as the local/passtest script used in the above example.

###############################################################################
# Further Information
#
#  See the snmpd.conf manual page, and the output of "snmpd -H".

rocommunity public 10.0.1.2
rocommunity public 127.0.0.1

syslocation "HYD, UM DataCenter"
syscontact admin@ndm.lk

-- INSERT --
```

5) Start SNMP service using the following command

```
root@mlb-dc1-centos9:~
[root@mlb-dc1-centos9 ~]# sudo systemctl status snmpd
○ snmpd.service - Simple Network Management Protocol (SNMP) Daemon.
     Loaded: loaded (/usr/lib/systemd/system/snmpd.service; disabled; preset: d>
     Active: inactive (dead)

[root@mlb-dc1-centos9 ~]# ^C
[root@mlb-dc1-centos9 ~]# vi /etc/snmp/snmpd.conf
[root@mlb-dc1-centos9 ~]# sudo systemctl start snmpd
[root@mlb-dc1-centos9 ~]# sudo systemctl status snmpd
● snmpd.service - Simple Network Management Protocol (SNMP) Daemon.
     Loaded: loaded (/usr/lib/systemd/system/snmpd.service; disabled; preset: d>
     Active: active (running) since Sat 2026-03-14 01:43:21 EDT; 30s ago
   Main PID: 3703 (snmpd)
      Tasks: 1 (limit: 22774)
     Memory: 9.9M (peak: 10.3M)
        CPU: 137ms
     CGroup: /system.slice/snmpd.service
             └─3703 /usr/sbin/snmpd -LS0-6d -f

Mar 14 01:43:21 mlb-dc1-centos9.csa.lk systemd[1]: Starting Simple Network Mana>
Mar 14 01:43:21 mlb-dc1-centos9.csa.lk snmpd[3703]: NET-SNMP version 5.9.1
Mar 14 01:43:21 mlb-dc1-centos9.csa.lk systemd[1]: Started Simple Network Manag>

[root@mlb-dc1-centos9 ~]# ^C
```

6) check out the -Tp flag:

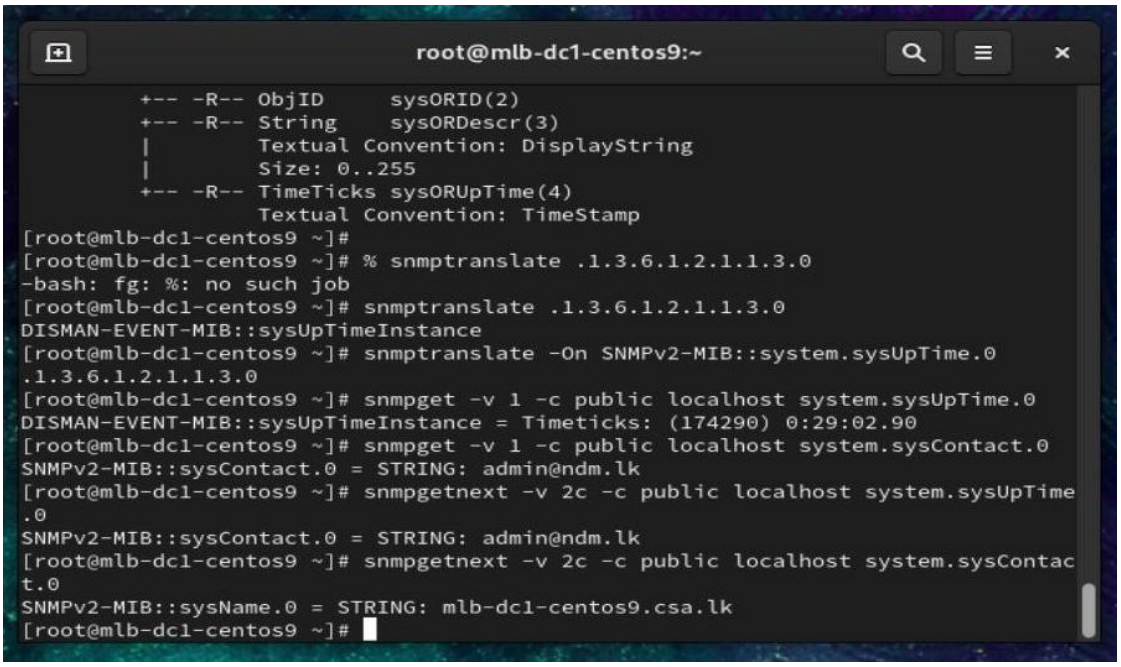
```
root@mlb-dc1-centos9:~
[root@mlb-dc1-centos9 ~]# snmptranslate -Tp
+--iso(1)
   |
   +--org(3)
      |
      +--dod(6)
         |
         +--internet(1)
            |
            +--directory(1)
            |
            +--mgmt(2)
            |  |
            |  +--mib-2(1)
            |     |
            |     +--system(1)
            |     |  |
            |     |  +-- -R-- String    sysDescr(1)
            |     |  |        Textual Convention: DisplayString
            |     |  |        Size: 0..255
            |     |  +-- -R-- ObjID     sysObjectID(2)
            |     |  +-- -R-- TimeTicks sysUpTime(3)
            |     |  |  |
            |     |  |  +--sysUpTimeInstance(0)
```

```
root@mlb-dc1-centos9:~
[root@mlb-dc1-centos9 ~]# snmptranslate -Tp -IR system
+--system(1)
   |
   +-- -R-- String    sysDescr(1)
   |        Textual Convention: DisplayString
   |        Size: 0..255
   +-- -R-- ObjID     sysObjectID(2)
   +-- -R-- TimeTicks sysUpTime(3)
   |  |
   |  +--sysUpTimeInstance(0)
   |
   +-- -RW- String    sysContact(4)
   |        Textual Convention: DisplayString
   |        Size: 0..255
   +-- -RW- String    sysName(5)
   |        Textual Convention: DisplayString
   |        Size: 0..255
   +-- -RW- String    sysLocation(6)
   |        Textual Convention: DisplayString
   |        Size: 0..255
   +-- -R-- INTEGER   sysServices(7)
   |        Range: 0..127
   +-- -R-- TimeTicks sysORLastChange(8)
   |        Textual Convention: TimeStamp
```

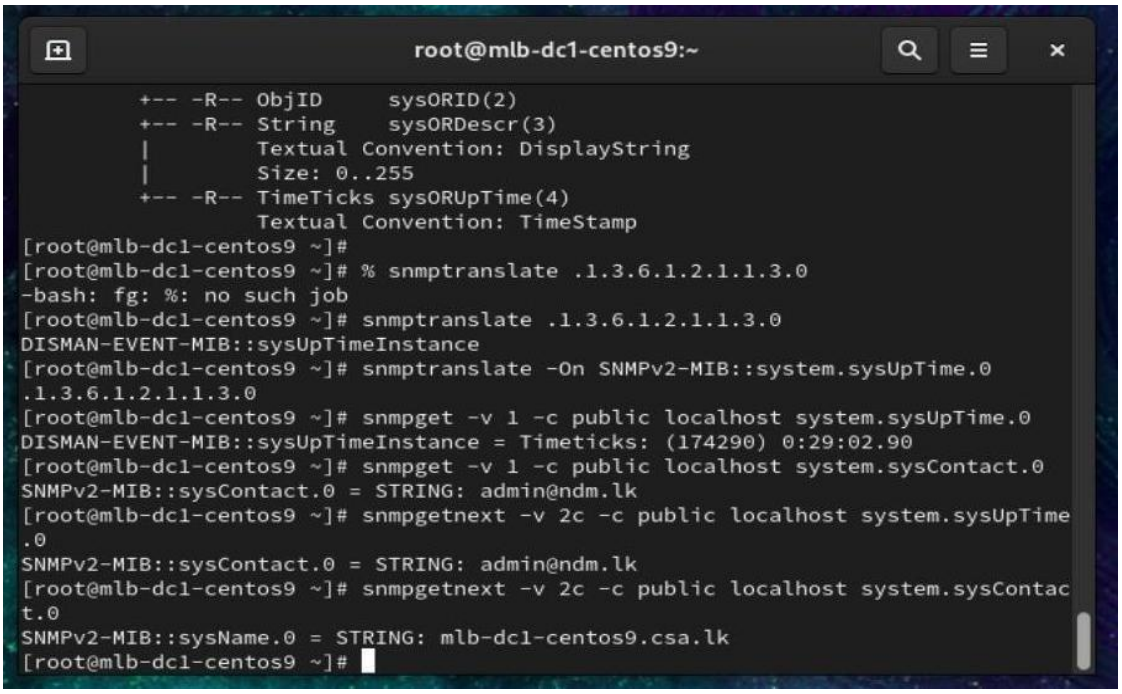
7) In its simplest form, it merely looks up an OID and spits it back out in textual form:

```
root@mlb-dc1-centos9:~
   +-- -R-- INTEGER   sysServices(7)
   |        Range: 0..127
   +-- -R-- TimeTicks sysORLastChange(8)
   |        Textual Convention: TimeStamp
   |
   +--sysORTable(9)
      |
      +--sysOREntry(1)
         |  Index: sysORIndex
         |
         +-- ---- INTEGER   sysORIndex(1)
         |        Range: 1..2147483647
         +-- -R-- ObjID     sysORID(2)
         +-- -R-- String    sysORDescr(3)
         |        Textual Convention: DisplayString
         |        Size: 0..255
         +-- -R-- TimeTicks sysORUpTime(4)
                  Textual Convention: TimeStamp
[root@mlb-dc1-centos9 ~]#
[root@mlb-dc1-centos9 ~]# % snmptranslate .1.3.6.1.2.1.1.3.0
-bash: fg: %: no such job
[root@mlb-dc1-centos9 ~]# snmptranslate .1.3.6.1.2.1.1.3.0
DISMAN-EVENT-MIB::sysUpTimeInstance
[root@mlb-dc1-centos9 ~]#
```

8) It can also translate into numerical results as well, by adding the -On flag to its options



```
root@mlb-dc1-centos9:~
        +-- -R-- ObjID     sysORID(2)
        +-- -R-- String    sysORDescr(3)
        |        Textual Convention: DisplayString
        |        Size: 0..255
        +-- -R-- TimeTicks sysORUpTime(4)
                 Textual Convention: TimeStamp
[root@mlb-dc1-centos9 ~]#
[root@mlb-dc1-centos9 ~]# % snmptranslate .1.3.6.1.2.1.1.3.0
-bash: fg: %: no such job
[root@mlb-dc1-centos9 ~]# snmptranslate .1.3.6.1.2.1.1.3.0
DISMAN-EVENT-MIB::sysUpTimeInstance
[root@mlb-dc1-centos9 ~]# snmptranslate -On SNMPv2-MIB::system.sysUpTime.0
.1.3.6.1.2.1.1.3.0
[root@mlb-dc1-centos9 ~]# snmpget -v 1 -c public localhost system.sysUpTime.0
DISMAN-EVENT-MIB::sysUpTimeInstance = Timeticks: (174290) 0:29:02.90
[root@mlb-dc1-centos9 ~]# snmpget -v 1 -c public localhost system.sysContact.0
SNMPv2-MIB::sysContact.0 = STRING: admin@ndm.lk
[root@mlb-dc1-centos9 ~]# snmpgetnext -v 2c -c public localhost system.sysUpTime
.0
SNMPv2-MIB::sysContact.0 = STRING: admin@ndm.lk
[root@mlb-dc1-centos9 ~]# snmpgetnext -v 2c -c public localhost system.sysContac
t.0
SNMPv2-MIB::sysName.0 = STRING: mlb-dc1-centos9.csa.lk
[root@mlb-dc1-centos9 ~]#
```



```
root@mlb-dc1-centos9:~
        +-- -R-- ObjID     sysORID(2)
        +-- -R-- String    sysORDescr(3)
        |        Textual Convention: DisplayString
        |        Size: 0..255
        +-- -R-- TimeTicks sysORUpTime(4)
                 Textual Convention: TimeStamp
[root@mlb-dc1-centos9 ~]#
[root@mlb-dc1-centos9 ~]# % snmptranslate .1.3.6.1.2.1.1.3.0
-bash: fg: %: no such job
[root@mlb-dc1-centos9 ~]# snmptranslate .1.3.6.1.2.1.1.3.0
DISMAN-EVENT-MIB::sysUpTimeInstance
[root@mlb-dc1-centos9 ~]# snmptranslate -On SNMPv2-MIB::system.sysUpTime.0
.1.3.6.1.2.1.1.3.0
[root@mlb-dc1-centos9 ~]# snmpget -v 1 -c public localhost system.sysUpTime.0
DISMAN-EVENT-MIB::sysUpTimeInstance = Timeticks: (174290) 0:29:02.90
[root@mlb-dc1-centos9 ~]# snmpget -v 1 -c public localhost system.sysContact.0
SNMPv2-MIB::sysContact.0 = STRING: admin@ndm.lk
[root@mlb-dc1-centos9 ~]# snmpgetnext -v 2c -c public localhost system.sysUpTime
.0
SNMPv2-MIB::sysContact.0 = STRING: admin@ndm.lk
[root@mlb-dc1-centos9 ~]# snmpgetnext -v 2c -c public localhost system.sysContac
t.0
SNMPv2-MIB::sysName.0 = STRING: mlb-dc1-centos9.csa.lk
[root@mlb-dc1-centos9 ~]#
```

Use snmpwalk

9) This command walks through the SNMP tree and shows all OID values.

```
root@mlb-dc1-centos9:~
e SNMP User-based Security Model.
SNMPv2-MIB::sysORDescr.4 = STRING: The MIB module for SNMPv2 entities
SNMPv2-MIB::sysORDescr.5 = STRING: View-based Access Control Model for SNMP.
SNMPv2-MIB::sysORDescr.6 = STRING: The MIB module for managing TCP implementatio
ns
SNMPv2-MIB::sysORDescr.7 = STRING: The MIB module for managing UDP implementatio
ns
SNMPv2-MIB::sysORDescr.8 = STRING: The MIB module for managing IP and ICMP imple
mentations
SNMPv2-MIB::sysORDescr.9 = STRING: The MIB modules for managing SNMP Notificatio
n, plus filtering.
SNMPv2-MIB::sysORDescr.10 = STRING: The MIB module for logging SNMP Notification
s.
SNMPv2-MIB::sysORUpTime.1 = Timeticks: (5) 0:00:00.05
SNMPv2-MIB::sysORUpTime.2 = Timeticks: (5) 0:00:00.05
SNMPv2-MIB::sysORUpTime.3 = Timeticks: (5) 0:00:00.05
SNMPv2-MIB::sysORUpTime.4 = Timeticks: (5) 0:00:00.05
SNMPv2-MIB::sysORUpTime.5 = Timeticks: (5) 0:00:00.05
SNMPv2-MIB::sysORUpTime.6 = Timeticks: (5) 0:00:00.05
SNMPv2-MIB::sysORUpTime.7 = Timeticks: (5) 0:00:00.05
SNMPv2-MIB::sysORUpTime.8 = Timeticks: (5) 0:00:00.05
SNMPv2-MIB::sysORUpTime.9 = Timeticks: (5) 0:00:00.05
SNMPv2-MIB::sysORUpTime.10 = Timeticks: (5) 0:00:00.05
[root@mlb-dc1-centos9 ~]#
```

Use snmptable

10) This command shows SNMP information in table format.

Use snmpset

11) This command changes SNMP values.

Use snmptrap

12) This command sends an SNMP trap message.

```
root@mlb-dc1-centos9:~
ns
SNMPv2-MIB::sysORDescr.7 = STRING: The MIB module for managing UDP implementatio
ns
SNMPv2-MIB::sysORDescr.8 = STRING: The MIB module for managing IP and ICMP imple
mentations
SNMPv2-MIB::sysORDescr.9 = STRING: The MIB modules for managing SNMP Notificatio
n, plus filtering.
SNMPv2-MIB::sysORDescr.10 = STRING: The MIB module for logging SNMP Notification
s.
SNMPv2-MIB::sysORUpTime.1 = Timeticks: (0) 0:00:00.00
SNMPv2-MIB::sysORUpTime.2 = Timeticks: (0) 0:00:00.00
SNMPv2-MIB::sysORUpTime.3 = Timeticks: (0) 0:00:00.00
SNMPv2-MIB::sysORUpTime.4 = Timeticks: (0) 0:00:00.00
SNMPv2-MIB::sysORUpTime.5 = Timeticks: (0) 0:00:00.00
SNMPv2-MIB::sysORUpTime.6 = Timeticks: (0) 0:00:00.00
SNMPv2-MIB::sysORUpTime.7 = Timeticks: (0) 0:00:00.00
SNMPv2-MIB::sysORUpTime.8 = Timeticks: (0) 0:00:00.00
SNMPv2-MIB::sysORUpTime.9 = Timeticks: (0) 0:00:00.00
SNMPv2-MIB::sysORUpTime.10 = Timeticks: (0) 0:00:00.00
[root@mlb-dc1-centos9 ~]# snmptable -v2c -c public localhost ifTable
IF-MIB::ifTable: No entries
[root@mlb-dc1-centos9 ~]# snmpwalk -v2c -c public localhost sysLocation
SNMPv2-MIB::sysLocation.0 = STRING: "HYD, UM DataCenter"
[root@mlb-dc1-centos9 ~]# snmptrap -v2c -c public localhost '' SNMPv2-MIB::coldS
tart
[root@mlb-dc1-centos9 ~]#
```